



Joints

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Joints

- Articulation between two or more bone
- Classified into

1. Fibrous

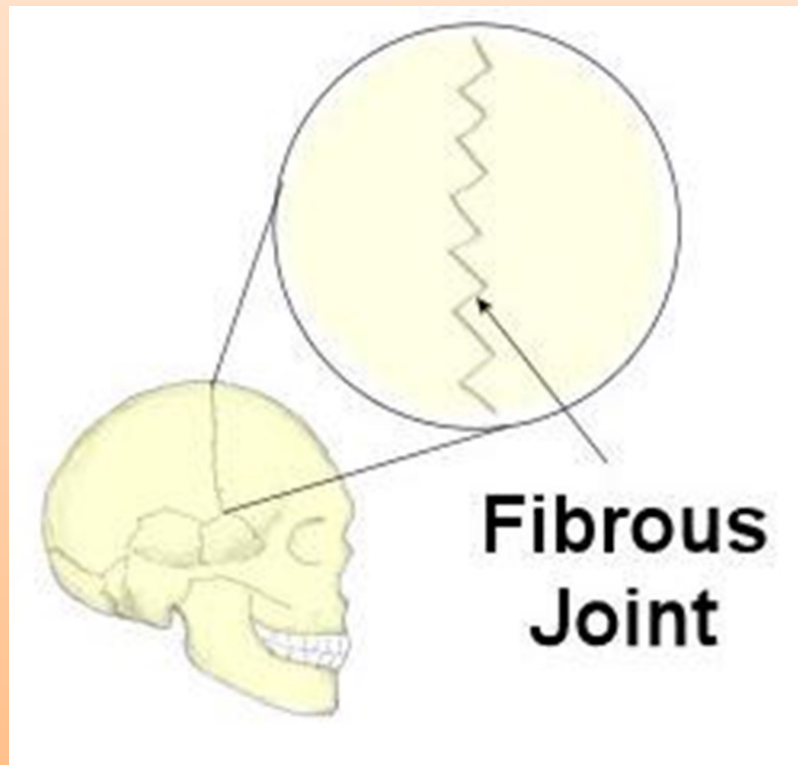
2. Cartilage and

3. Synovial joint

❖ *There are 186 joints in the body*

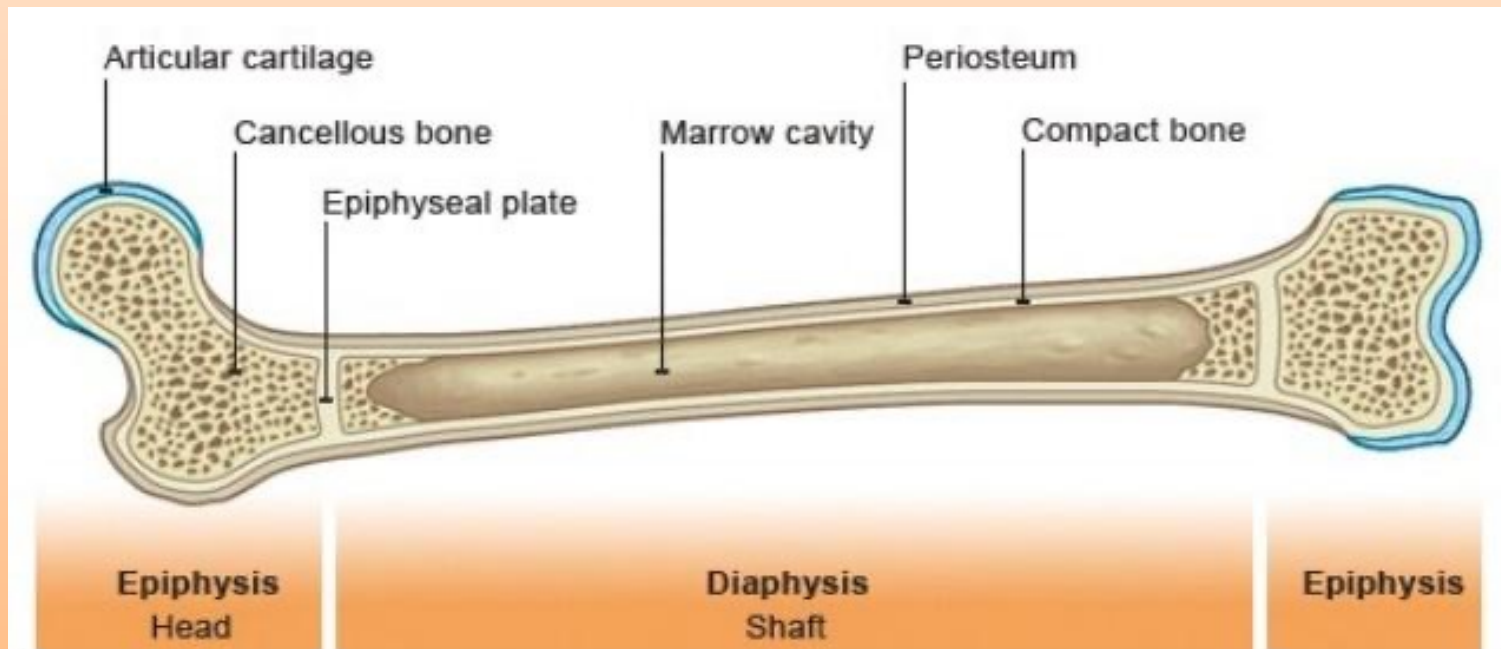
Fibrous joint

The articulating surfaces of the bones are joined by fibrous tissue and thus very little or no movement is possible. Ex. Sutures of skull.



Cartilaginous joint

- **Two types**
- **Primary cartilaginous joint** : the bone are articulated by hyaline cartilage such as epiphysial plate of long bone



Secondary cartilaginous joint

- The bones are connected by fibrous cartilage

- Such as intervertebral disc

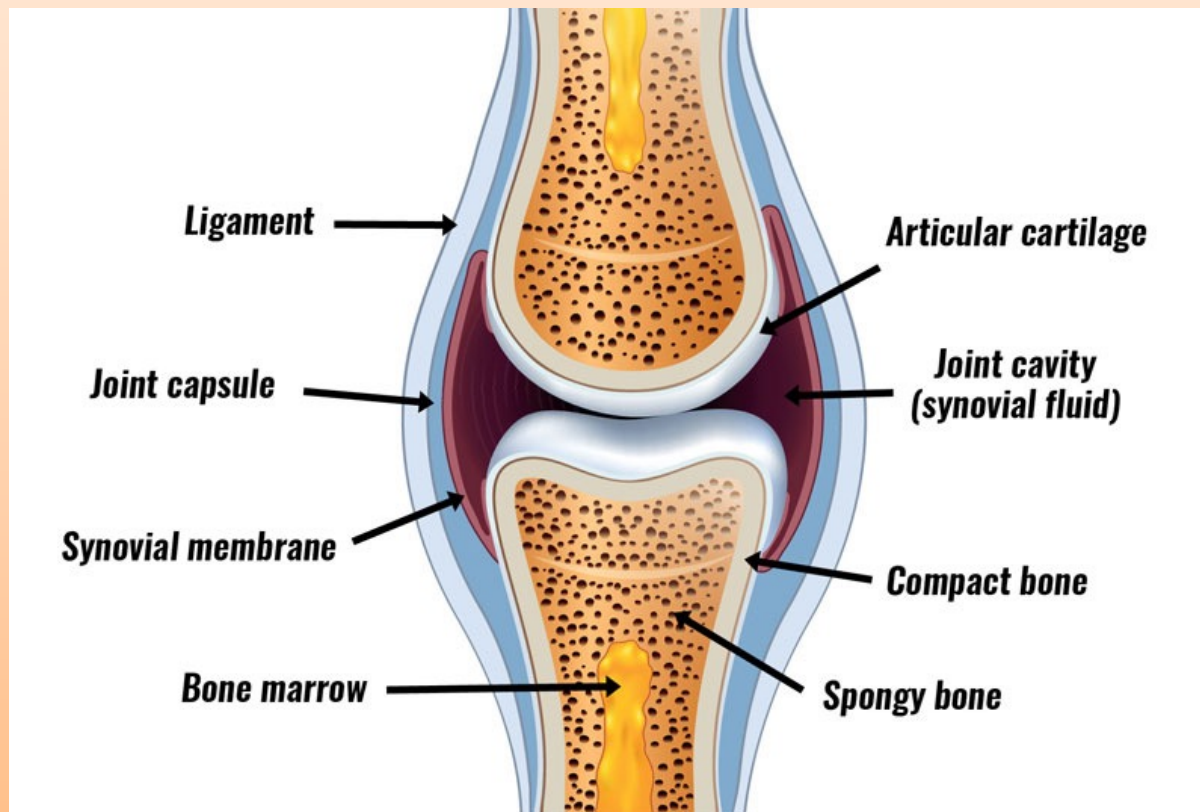
Which connects the body

Of two vertebrae



Synovial joint

- Are true mobile joints



Synovial joints.

- The articular surfaces of the bones are covered by a thin layer of *hyaline cartilage* separated by a *joint cavity*.
- The margins of the articular surfaces connected which each other by strong fibrous tissue called *capsule*
- The capsule lined internally by *synovial membrane*

- The articular surfaces are lubricated by a viscous fluid called *synovial fluid*, which is produced by synovial membrane.
- The capsule is strengthened by *ligaments* which unites the bones together.
- Some ligaments lie within the capsule, some others lie outside it.

Types of synovial joints

- According to the shape of articulating bones, and the type of movement the synovial joints are classified into the following:
 - 1- **Plane joints (Gliding joint)**: the articular surfaces are flat and this permits the bones to slide over one another ex. *Acromioclavicular joint*.
 - 2- **Hinge joints (Uniaxial)**: only flexion and extension are possible. Ex. *Elbow, knee, and ankle joints*.

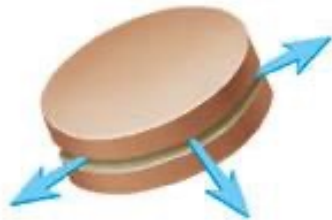
- 3- **Pivot joints (Uniaxial)**: : in which a central bony pivot is surrounded by a bony-ligamentous ring and rotation is the only movement possible. Ex. *Atlanto-axial joint and superior radioulnar joints*.
- 4- **Condyloid joints (Bi-axial)** : in which there is two distinct convex surfaces articulate with two concave surfaces. Flexion, extension, abduction, adduction together with a small amount of rotation are possible, ex. *Metacarpophalangeal joints*.

5- Ellipsoid joints (Bi-axial) : here an elliptical convex articular surface fits into an elliptical concave articular surface. Flexion, extension, abduction and adduction are possible with no rotation. Ex. *Wrist joint*.

6- Saddle joint (Bi-axial) : in saddle joints, the articular surfaces are reciprocally concavo-convex and resemble a saddle on a horse's back. Flexion, extension, abduction, adduction and rotation are possible . Ex. *Carpometacarpal joint of the thumb*.

7- Ball and socket joints (Multi-axial) : here a ball shaped convex surface fits into a socket like concavity. It permits free movement. (flexion, extension, abduction, adduction, medial rotation, lateral rotation and circumduction)ex. *Shoulder and hip joints*.

Types of synovial joints



Plane Joint



Saddle Joint



Hinge Joint



Pivot Joint



Ball-and-Socket Joint



Ellipsoid Joint

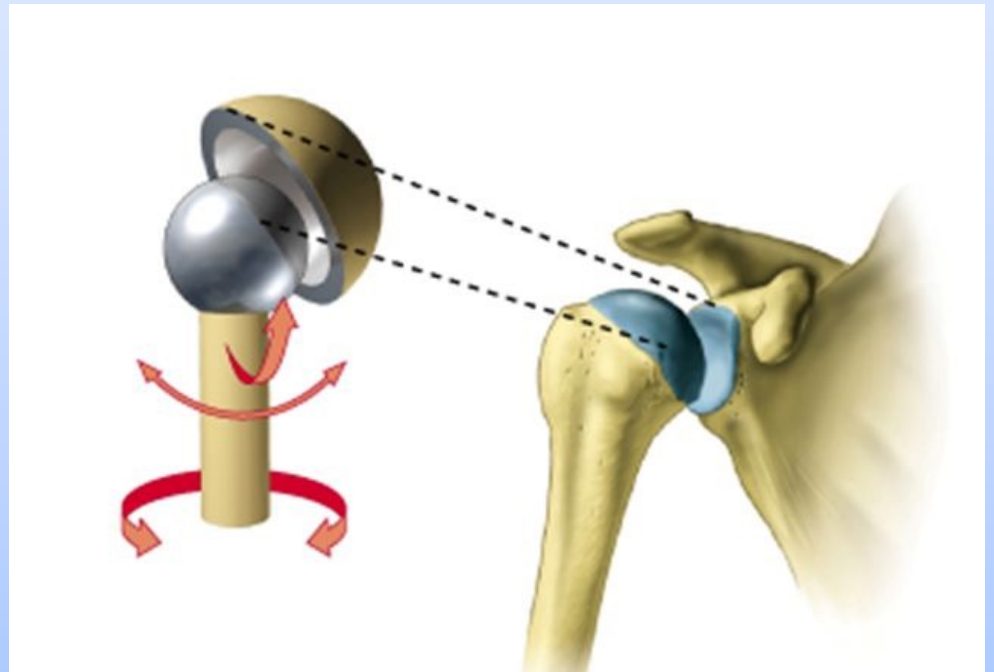


Objectives

❖ *Type*

❖ *Articulation*

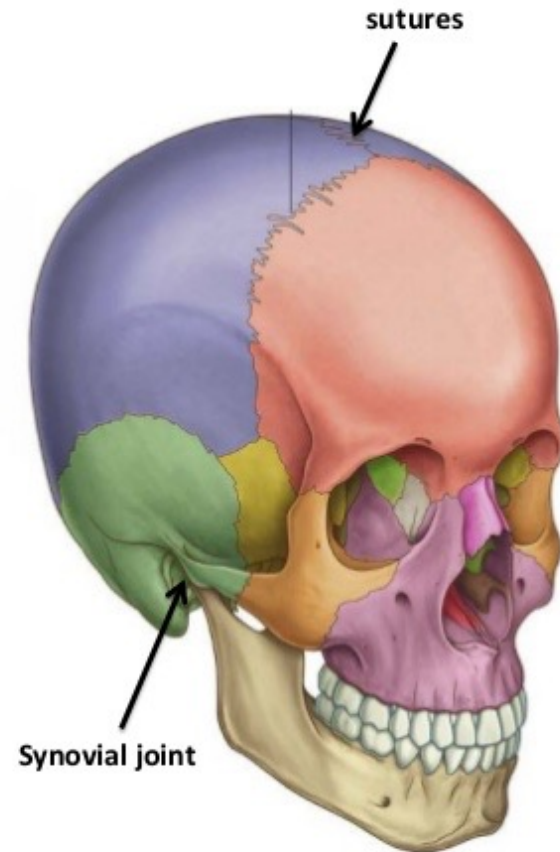
❖ *Movements*

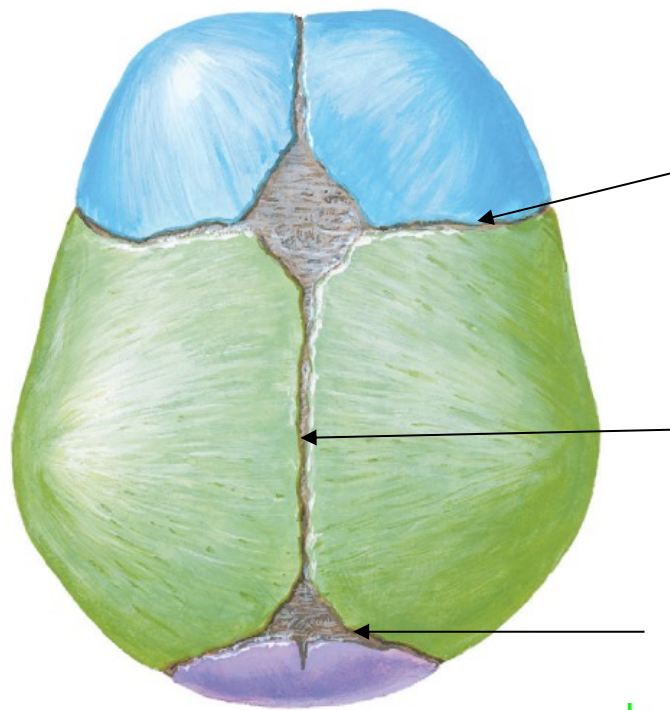


Joints of the head and neck

Joints of skull

- Most of the joint of skull are **sutures** (fibrous joint).
- The largest are the **temporomandibular joints** between the lower jaw (mandible) and the temporal bone.

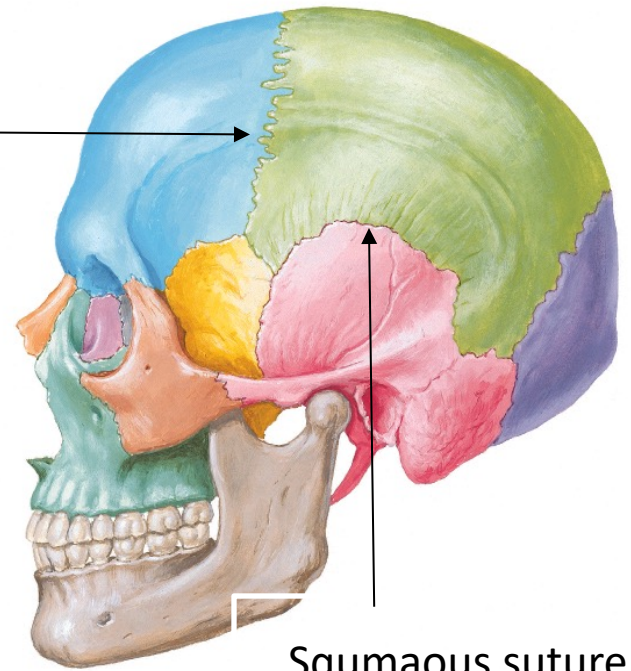




Coronal suture

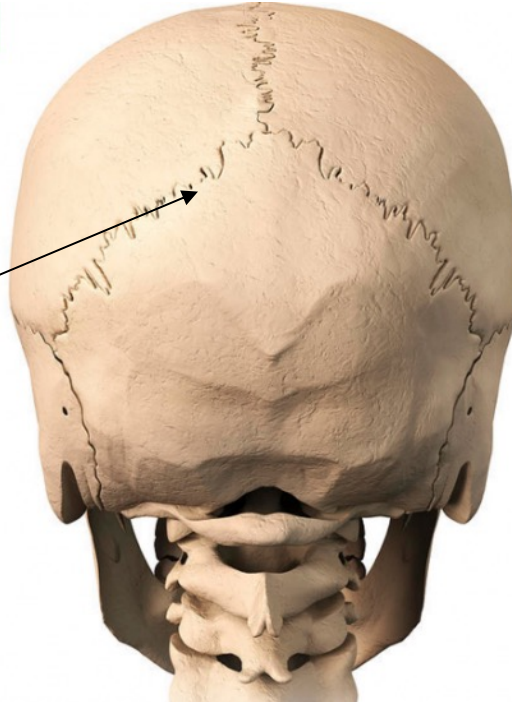
Sagittal suture

Lambdoid suture



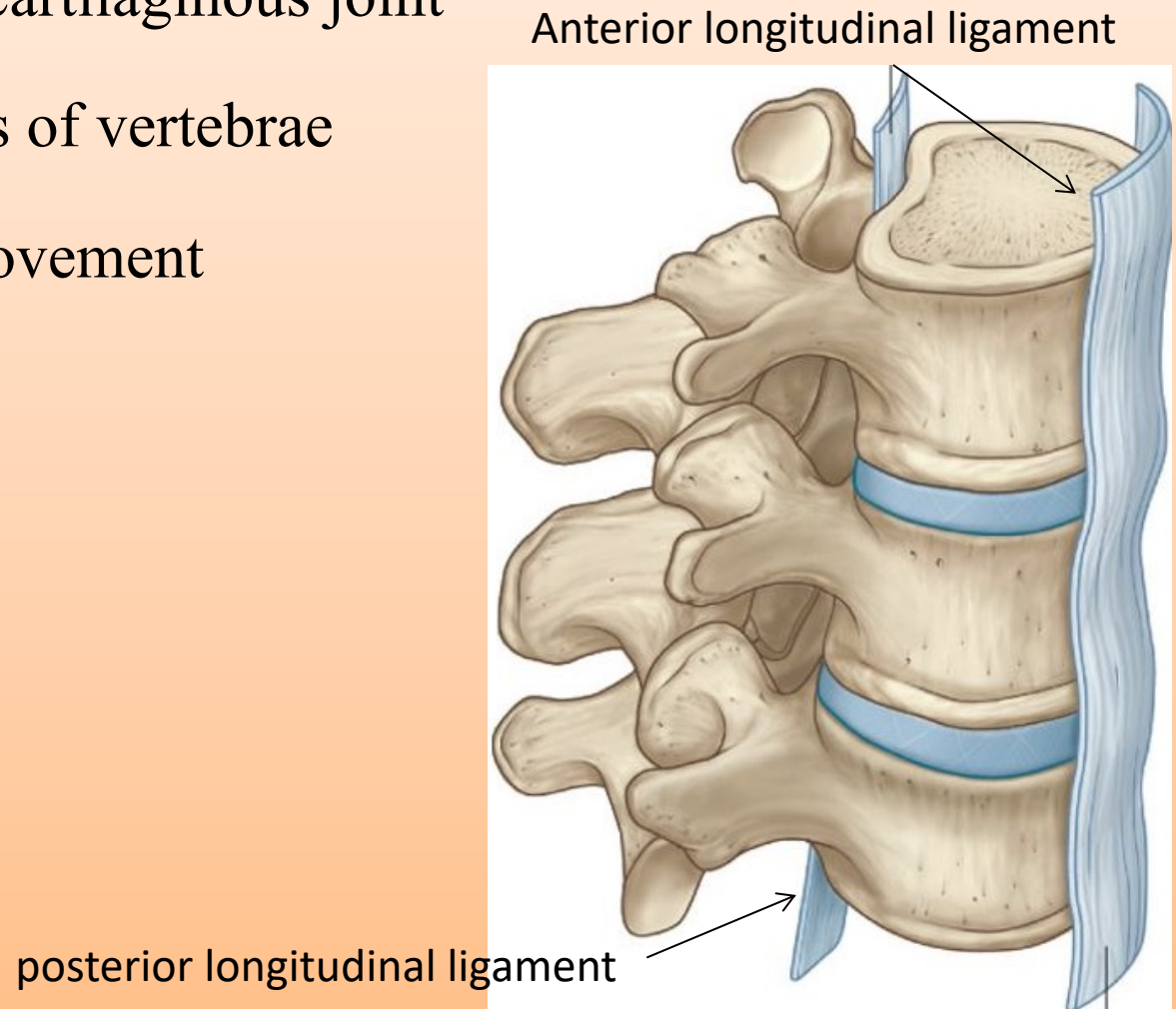
Squamous suture

Lambdoid suture



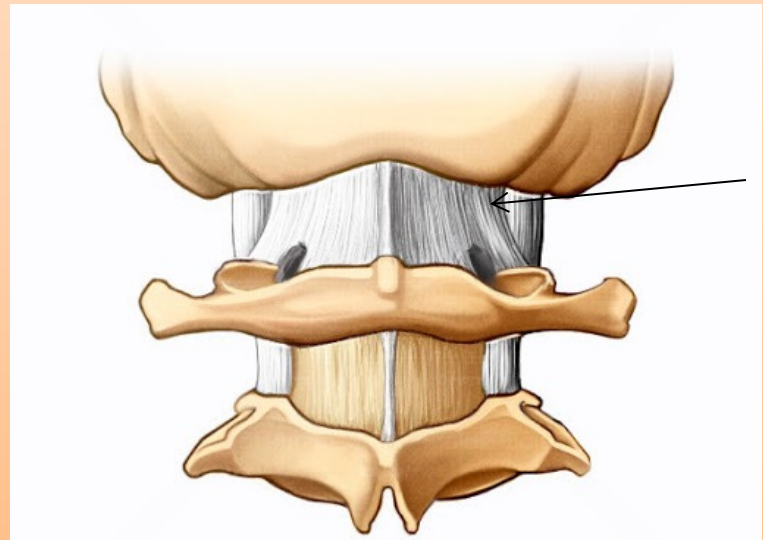
Joints between two vertebral bodies

- Type: Is secondary cartilaginous joint
- Articulation: bodies of vertebrae
- Movements: slid movement



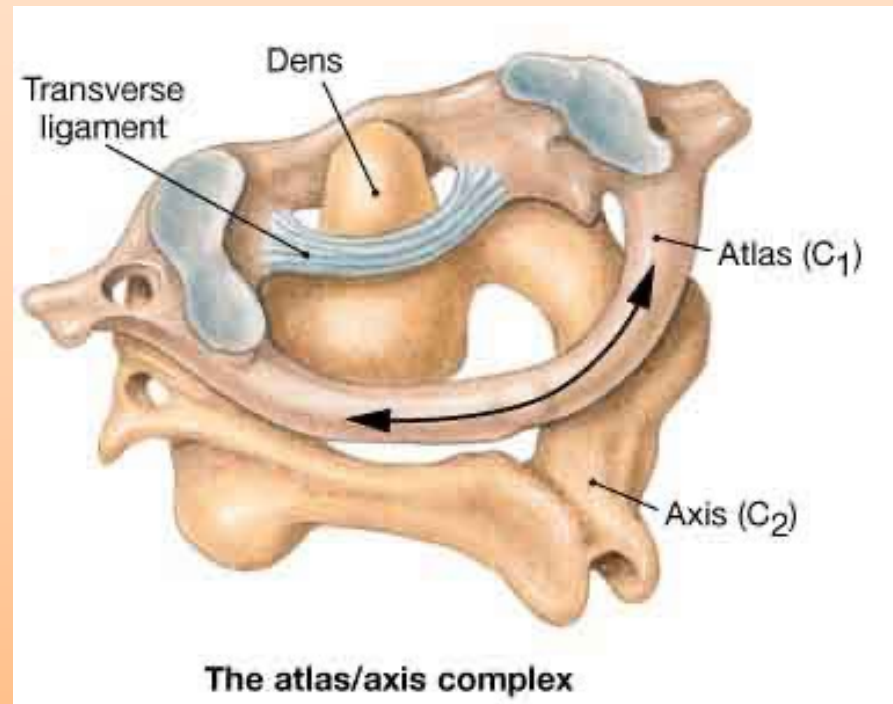
Atlanto-occipital joint

- Are synovial joints that are formed between the occipital condyles superiorly and the facets on the superior surfaces of the lateral masses of the atlas inferiorly
- **Movements:** Flexion, extension, and lateral flexion. No rotation is possible
- *Biaxial joint*



Atlantoaxial Joints

- ❖ **Type** : Pivote type of synovial joints
- ❖ **Articulation**: between the *odontoid process* and the *anterior arch of the atlas*, and the other two are *between the lateral masses of the bones*
- ❖ **Movement**: *Rotation*



Joints of the Upper Limb

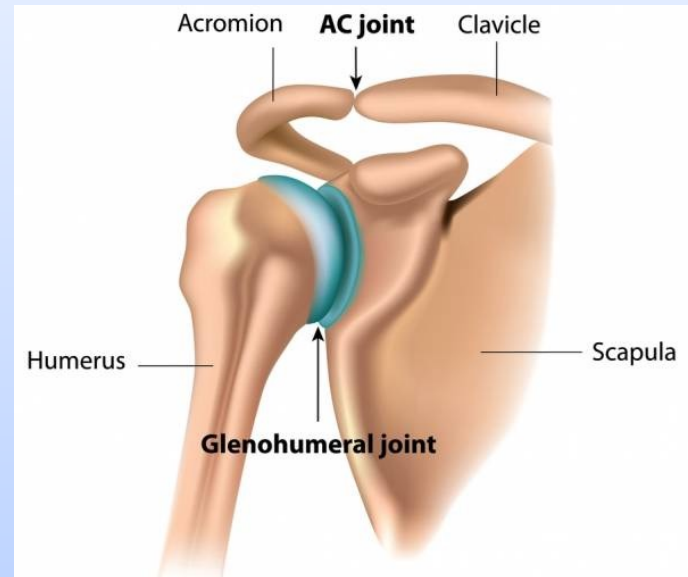
Acromioclavicular joint(ACJ)

➤ **Articulation:** This occurs between the acromion of the scapula and the lateral end of the clavicle.

➤ **Type:** Synovial plane joint

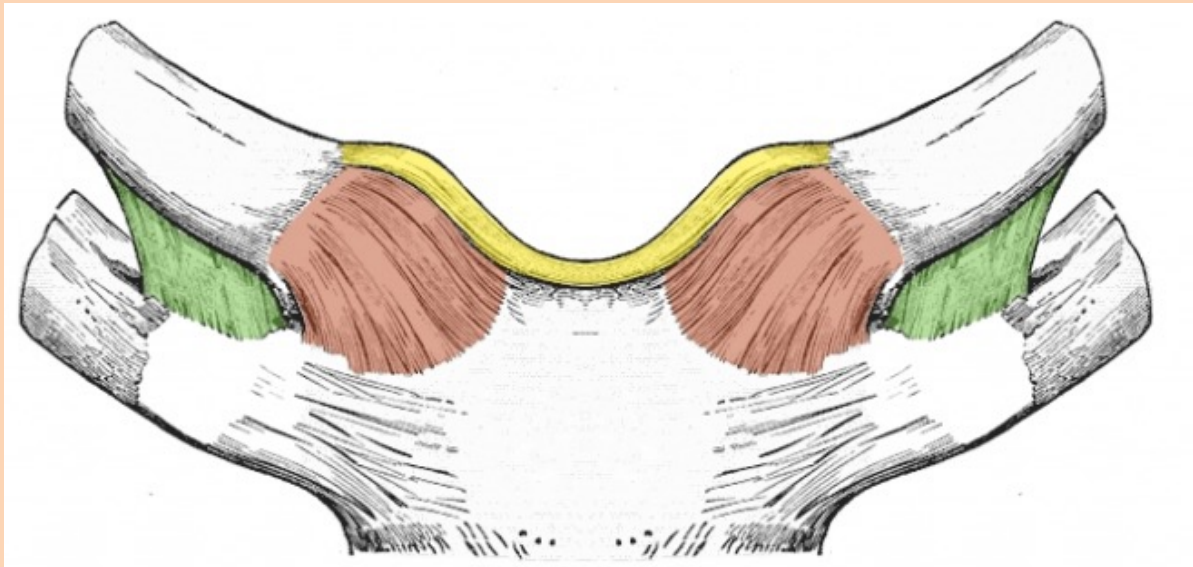
- **Movements**

A gliding movement



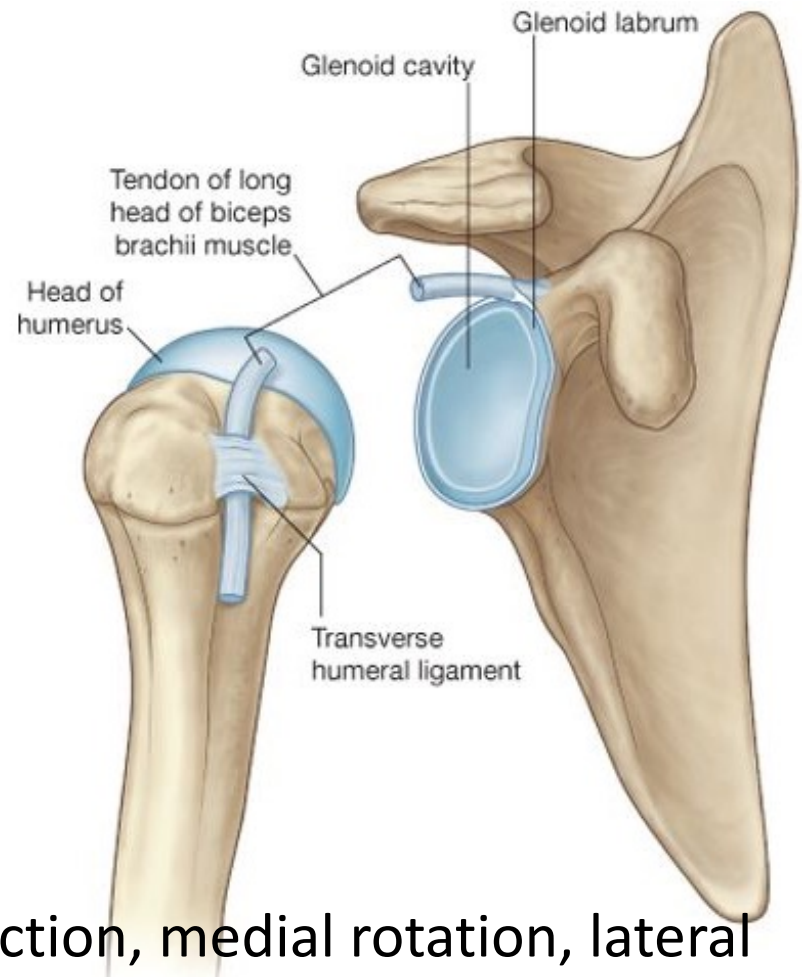
Sternoclavicular Joint

- **Articulation:** occurs between the sternal end of the clavicle, the manubrium sterni, and the first costal cartilage
- **Type:** Synovial double-plane joint



Shoulder joint Glenohumeral joint

- Type is a synovial ball and socket (multi-axial)
- Articulation between the head of the humerus and the glenoid cavity of the scapula



Movements

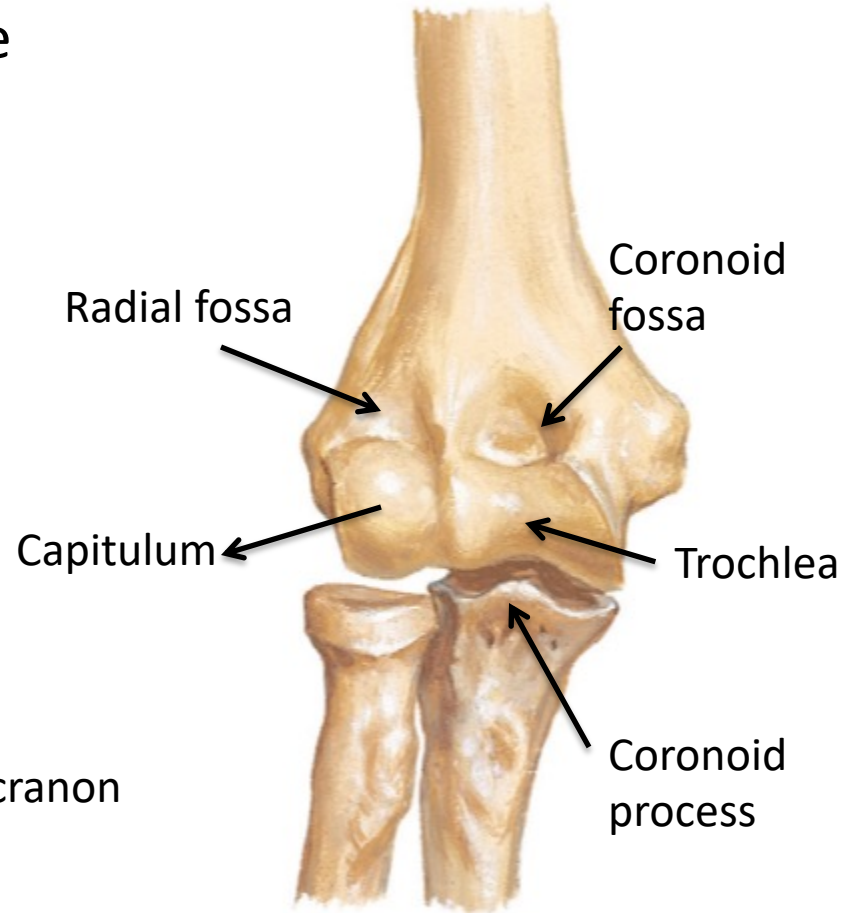
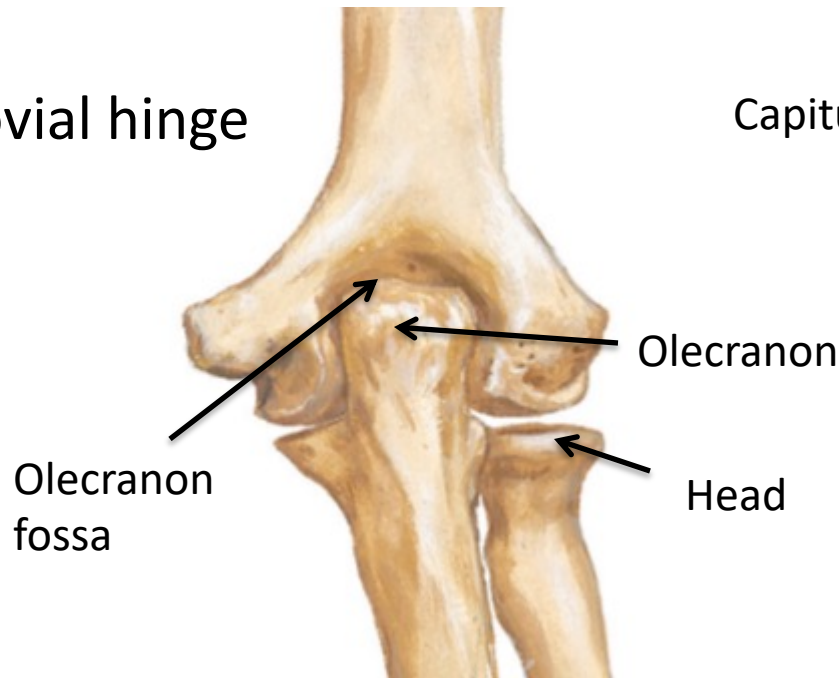
Flexion, extension, abduction, adduction, medial rotation, lateral rotation, and circumduction are possible.

Elbow joint

Articulation: This occurs between the trochlea and capitulum of the humerus and the trochlear notch of the ulna and the head of the radius

It permits only flexion and extension

Type: Synovial hinge joint



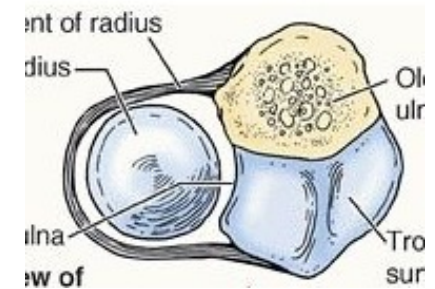
The radioulnar joints

- **The two ends** (upper and lower ends) of the radius and ulna are jointed to each other *at the superior and inferior radioulnar joints*
- **The shaft** of the two bones are united by the *interosseous membrane* (some time called middle radioulnar joint)



Superior radioulnar joint

- **Articulation:** The head of the radius with the radial notch of the ulna. The radial head is held in position by the anular ligament of the radius
- **Type:** Synovial pivot joint
- *Possible movements are pronation and supination.*



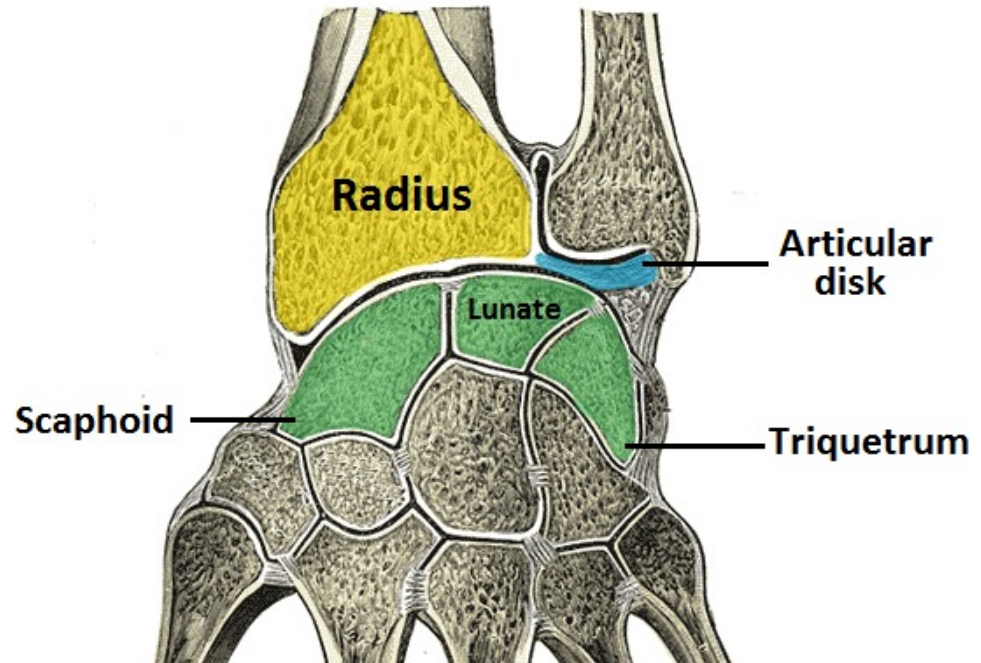
Distal Radio-Ulnar Joint

- **Articulation:** Between the rounded head of the ulna and the ulnar notch on the radius
- **Type:** Synovial pivot joint
- *Possible movements are pronation and supination.*



Wrist joint (Radiocarpal Joint)

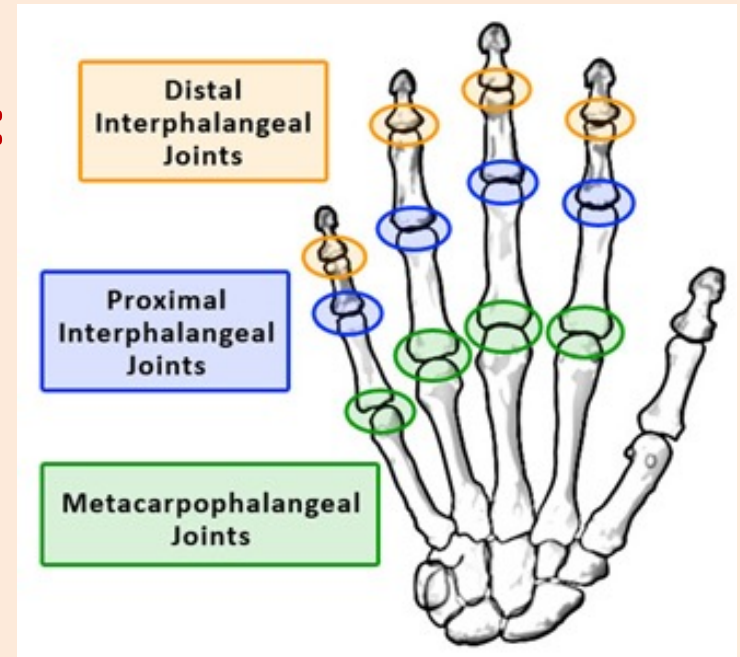
- **Articulation:** Is between the distal end of radius and the articular disc of ulna above, and the scaphoid, lunate and triquetral bones below.
- **Type:** it is a synovial ellipsoid joint).
- *It allows flexion, extension, abduction and adduction, with no rotation.*



Joints of the Hand and Fingers

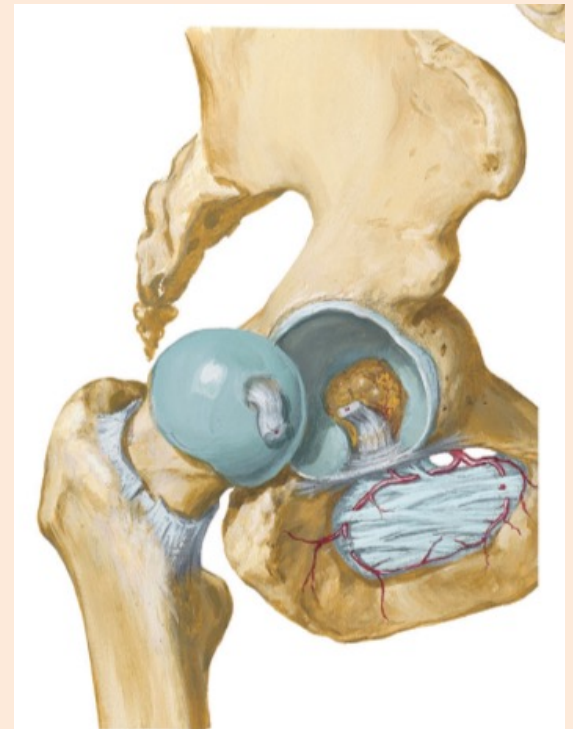
hand joints include

- ❖ **Intercarpal joints:** Plane type,
- ❖ **Carpometacarpal joints:** Plane type except the thumb which is saddle,
- ❖ **Metacarpophalangeal joints:** are biaxial
- ❖ **Interphalangeal joints:** are hinge or uniaxial.



Hip joint

- **Type:** Is a synovial ball and socket joint
- **Articulation:** between the head of the femur and acetabulum
- **Movements:** flexion, Extension, abduction, adduction, lateral and medial rotation ,and circumduction



Knee joint

- Is the most complex joint of the body
- **Articulation** : Between two rounded condyles of femur above and two condyles of the tibia and their cartilaginous menisci
- **Types**: Between the femur and tibia is a synovial hinge variety, Between the patella and femur is synovial plane gliding variety



Proximal tibiofibular joint

- **Type is** a plane type of synovial joint
- **Articulation** between the lateral condyle of the tibia and head of the fibula



Distal Tibiofibular Joint

- **Articulation**
 - Articulation is between the fibular notch of the tibia and the lower end of the fibula
- **Type**
 - The distal tibiofibular joint is a fibrous joint.



Ankle Joint

- **Articulation**

- Articulation is between the lower end of the tibia, the two malleoli, and the body of the talus

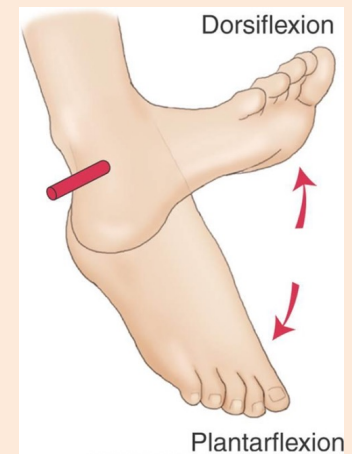
- **Type**

- The ankle is a synovial hinge joint



Movement:

Dorsiflexion (toes pointing upward) and **plantar flexion** (toes pointing downward) are possible. The movements of inversion and eversion take place at the tarsal joints and **not at the ankle joint**.



THANK
YOU